CS182: Introduction to Machine Learning
Lecture 07: Linear Discrimination

Worked Solutions and Marking Guide

Scope: Linear_Discrimination.pdf. Independent practice material based on the official 2026 Fall lecture.

Total: 100 marks. **Suggested time:** 120 minutes.

Section	Questions	Each	Marks
A. Misconceptions: True/False	12	1	12
B. Multiple response	12	2	24
C. Proofs and formula derivations	4	10	40
D. Applications	2	12	24
Total	30		100

Scoring. In Section B each item has exactly four options, of which one, two, or three are correct. The complete correct set earns 2 marks; a nonempty proper subset earns 1 mark; any incorrect option or no selection earns 0. T/F answers earn 1 mark only when correct. Show reasoning in Sections C and D; part marks are available for valid intermediate work. Equivalent correct methods receive credit.

Instructions. Answer in English. Use natural logarithms. Exact values and justified approximations are accepted. All vectors are columns unless stated otherwise. Use local assumptions, label conventions, and tie rules.

Quick key — Section A

1	2	3	4	5	6
F	F	T	T	T	T

7	8	9	10	11	12
F	T	F	T	F	F

Quick key — Section B

13	14	15	16	17	18
AB	D	C	ABD	BD	AD

19	20	21	22	23	24
BC	A	AC	CD	ABD	BCD

Section A. True/False — Explanations

1. False. Each class region is an intersection of score-comparison half-spaces and is convex. This statement concerns affine-score argmax, not arbitrary voting combinations.
Source: PDF p.18.

2. False. This requires $w \neq 0$. If $w = 0$, the zero set is either the whole space or empty, depending on the intercept.
Source: PDF pp.11-15; supplementary degenerate case.

3. True. The score sign is unchanged; sigmoid is strictly monotone but not invariant to multiplication of its argument. Classification geometry and probability calibration have different scale behavior.
Source: PDF pp.11,26,36; supplementary scale argument.

4. True. The parameter update is outside the full sample loop. Updating after each sample would be a different, sequential optimization procedure.
Source: PDF pp.35,44.

5. True. For finite real a, both the exponential and the denominator are positive, and $0 < 1/(1+e^{-a}) < 1$. Values zero or one are limits rather than finite-logit outputs.
Source: PDF pp.26,36-37.

6. True. For a nontrue class j, the error is $y_j - 0$, generally nonzero because its score competes in the shared normalization.
Source: PDF p.43.

7. False. Independent sigmoids do not share a denominator. Softmax imposes a joint normalization across all classes; independent binary probabilities need not sum to one.
Source: PDF pp.17-18,38-41; supplementary comparison.

8. True. The common exponential factor cancels from numerator and denominator at each input. This also creates parameter non-identifiability in the symmetric parameterization.
Source: PDF pp.40-41; supplementary identifiability argument.

9. False. Gaussian LDA estimates class-conditional distributions and priors, whereas logistic regression optimizes a conditional Bernoulli likelihood. The same form of boundary does not identify the training objective.
Source: PDF pp.4-6,22-31.

10. True. The pairwise score constrains the relative classes only; samples from a third class have no required sign for this subproblem.
Source: PDF pp.19-20.

11. False. The assumed affine quantity is the conditional log-odds. Gaussian shared-covariance distributions imply such a model, but they are not necessary to fit it; discrete features are allowed.
Source: PDF pp.29-31,47.

12. False. Along a separating direction the parameter norm can diverge while cross-entropy decreases toward zero without attaining zero at any finite vector. Convexity does not ensure attainment.
Source: PDF pp.32,36-37; supplementary existence analysis.

Section B. Multiple Response — Explanations

13. Correct set: A, B.

(A) **Correct.** Averaging divides the full gradient by four.

(B) **Correct.** Then the scale factor in the learning rate cancels the factor 1/4 in the gradient.

(C) **Incorrect.** Initial predictions are 1/2, but the weighted label residuals generally do not cancel.

(D) **Incorrect.** Sequential updates evaluate later gradients at changed weights.

Source: PDF pp.32-35; supplementary sum/mean comparison.

14. Correct set: D.

(A) **Incorrect.** A common shift preserves all pairwise score differences.

(B) **Incorrect.** The specified rule chooses the largest score and includes no reject action.

(C) **Incorrect.** Argmax on the raw scores is already a fully specified classification rule.

(D) **Correct.** Minus one is the largest of the three scores. Scores need not be positive to win an argmax.

Source: PDF pp.17-18.

15. Correct set: C.

(A) **Incorrect.** The risk is $2(0.55) = 1.10$.

(B) **Incorrect.** The cost-sensitive threshold is 0.6; at this point C_2 has lower risk.

(C) **Correct.** Compare $3(1 - y)$ against $2y$.

(D) **Incorrect.** The threshold is $\log((3/5)/(2/5)) = \log(3/2)$.

Source: PDF pp.24,30; supplementary application of lecture 05 loss rule.

16. Correct set: A, B, D.

(A) **Correct.** The gradient of the full quadratic penalty is $\lambda\theta$.

(B) **Correct.** This is a model-selection choice; the final test set should remain independent.

(C) **Incorrect.** L2 shrinkage does not in general produce exact sparsity.

(D) **Correct.** The positive quadratic penalty is coercive and makes the convex loss strongly convex.

Source: PDF p.37; supplementary connection to regularization.

Section B. Multiple Response — Explanations

17. Correct set: B, D.

(A) **Incorrect.** The derivative is strictly positive for every finite logit.

(B) **Correct.** $\sigma'(a) = \sigma(a)(1 - \sigma(a))$ is maximized at probability $1/2$.

(C) **Incorrect.** For example, take both arguments zero: the two sides are $1/2$ and one.

(D) **Correct.** The odds are nine, so the probability is $9/(1 + 9)$.

Source: PDF pp.26,33; supplementary maximum calculation.

18. Correct set: A, D.

(A) **Correct.** It equals $y - r = 1/4 - 1$.

(B) **Incorrect.** The gradient is negative, so a descent step increases the intercept.

(C) **Incorrect.** Multiply the error $-3/4$ by each feature; the second entry is $+3/2$.

(D) **Correct.** For a positive label, the loss is $-\log y$.

Source: PDF pp.31-34.

19. Correct set: B, C.

(A) **Incorrect.** It shows sufficiency of a shared-covariance Gaussian model, not necessity.

(B) **Correct.** The logit slope is $(1 - (-1))/2 = 1$, and the intercept is zero.

(C) **Correct.** The prior contribution is the additive log prior ratio.

(D) **Incorrect.** An unequal-variance log density ratio generally retains a quadratic term.

Source: PDF pp.22-30,48.

20. Correct set: A.

(A) **Correct.** For different indices the derivative is $-y_2 y_1$.

(B) **Incorrect.** Every column sums to zero, since the total probability is the constant one.

(C) **Incorrect.** The derivative is $0.5(1 - 0.5) = 0.25$.

(D) **Incorrect.** The common denominator gives off-diagonal derivatives $-y_i y_j$.

Source: PDF p.43.

Section B. Multiple Response — Explanations

21. Correct set: A, C.

(A) **Correct.** It becomes $g = z - 1$, an affine function of the new feature.

(B) **Incorrect.** Linearity in coefficients does not force a linear boundary in the original input.

(C) **Correct.** The zero set is $x_1^2 + x_2^2 = 1$.

(D) **Incorrect.** A fixed basis-function discriminant does not itself impose Gaussian input densities.

Source: PDF pp.8-9,48; supplementary example.

22. Correct set: C, D.

(A) **Incorrect.** Ordinary distance is nonnegative; its value is $|-5|/5 = 1$.

(B) **Incorrect.** The score must be divided by $\|w\| = 5$, giving distance $3/5$.

(C) **Correct.** $w^\mathsf{T} x + w_0 = 8 - 5 = 3$.

(D) **Correct.** Subtract $(3/25)(4, -3)^\mathsf{T}$ from $(2, 0)^\mathsf{T}$.

Source: PDF pp.13-15; supplementary numerical example.

23. Correct set: A, B, D.

(A) **Correct.** The exponentials are $(1, 3, 2)$, whose sum is six.

(B) **Correct.** The same exponential factor cancels from the softmax.

(C) **Incorrect.** Multiclass argmax needs no universal $1/2$ threshold; rejection requires a separately specified rule.

(D) **Correct.** The common denominator cancels in the probability ratio.

Source: PDF pp.38-41.

24. Correct set: B, C, D.

(A) **Incorrect.** It adds a common affine score, which cancels.

(B) **Correct.** Three remaining classes each need two weights and one intercept.

(C) **Correct.** There are four rows, each with three coefficients.

(D) **Correct.** With sample rows, it is $(Y - R)^\mathsf{T} \widetilde{X}$.

Source: PDF pp.38-44; supplementary matrix and identifiability notation.

Section C. Proofs and Formula Derivations

25. Derive the geometry rather than treating the score as a distance [10 marks]

(a) [2 marks] For boundary points u, v, subtract $w^\mathsf{T}u+b=0$ and $w^\mathsf{T}v+b=0$. This gives $w^\mathsf{T}(u-v)=0$. Every displacement within the affine hyperplane is of this form, so w is perpendicular to it.

(b) [4 marks] Substitution gives $g(x) = g(x_p) + rw^\mathsf{T}w/\|w\|_2 = r\|w\|_2$. Hence $r = g(x)/\|w\|_2$, and rearranging the decomposition gives

$$x_p = x - \frac{w^\mathsf{T}x + b}{\|w\|_2^2}w.$$

The ordinary distance is $|r|$, not r. Substituting the projection in the boundary equation yields zero and verifies the formula.

(c) [4 marks] The new score is $cg(x)$, so its zero set and sign are unchanged. Its norm is $c\|w\|$, so $cg(x)/(c\|w\|) = g(x)/\|w\|$. The probability becomes $\sigma(cg(x))$, generally different when $c \neq 1$ and $g(x) \neq 0$. If $w = 0$, the score is constant and the zero set is all of space when $b = 0$, or empty otherwise; the distance formula is undefined.

Marking: use the displayed subpart allocations; award credit for the logical or numerical intermediate steps.

Source: PDF pp.11-15; supplementary projection and scale proof.

Section C. Proofs and Formula Derivations

26. Derive the LDA-to-logistic bridge with its assumptions [10 marks]

(a) [5 marks] Bayes rule cancels the common evidence. The Gaussian normalizers also cancel because covariance is shared. Expanding both quadratic exponents gives

$$\log \frac{P(C_1 \mid x)}{P(C_2 \mid x)} = (\mu_1 - \mu_2)^\top \Sigma^{-1} x - \frac{1}{2}(\mu_1^\top \Sigma^{-1} \mu_1 - \mu_2^\top \Sigma^{-1} \mu_2) + \log \frac{\pi_1}{\pi_2}.$$

The common $x^\top \Sigma^{-1} x$ cancels and symmetry combines the cross terms. Thus $w = \Sigma^{-1}(\mu_1 - \mu_2)$ and $b = -\frac{1}{2}(\mu_1^\top \Sigma^{-1} \mu_1 - \mu_2^\top \Sigma^{-1} \mu_2) + \log(\pi_1/\pi_2)$.

(b) [2 marks] Exponentiating gives $e^a(1-y) = y$, so $y = e^a/(1+e^a) = \sigma(a)$. Since sigmoid is increasing and $\sigma(0) = 1/2$, $y > 1/2$ is equivalent to $a > 0$.

(c) [3 marks] With unequal covariances, the difference contains $-\frac{1}{2}x^\top(\Sigma_1^{-1} - \Sigma_2^{-1})x$, generally quadratic. Shared Gaussian covariances are a sufficient generative explanation of affine posterior log-odds. Logistic regression instead assumes that conditional log-odds are affine and fits the conditional label likelihood directly, without requiring Gaussian class densities.

Marking: use the displayed subpart allocations; award credit for the logical or numerical intermediate steps.

Source: PDF pp.22-30.

Section C. Proofs and Formula Derivations

27. Derive Bernoulli cross-entropy, its gradient, and its limits [10 marks]

(a) [3 marks] Each label has probability $(y^t)^{r^t}(1-y^t)^{1-r^t}$. Conditional independence gives

$$L(\theta) = \prod_t (y^t)^{r^t}(1-y^t)^{1-r^t}, \quad E(\theta) = -\sum_t [r^t \log y^t + (1-r^t)\log(1-y^t)].$$

This is binary cross-entropy; the inputs are held fixed during conditional fitting.

(b) [5 marks] For one sample, $dy/da = y(1-y)$ and $\partial \ell/\partial y = -r/y + (1-r)/(1-y)$. Their product is $-r(1-y) + (1-r)y = y - r$. Because $\partial a/\partial \theta = \widetilde{x}$, summation gives the gradient. Differentiating again yields

$$H = \sum_t y^t(1-y^t)\widetilde{x}^t(\widetilde{x}^t)^\mathsf{T}.$$

For any v, $v^\mathsf{T} H v = \sum_t y^t(1-y^t)(v^\mathsf{T}\widetilde{x}^t)^2 \geq 0$, establishing convexity. The descent increment is minus this gradient.

(c) [2 marks] Both samples have correct-label probability $\sigma(2c)$, so $E(c) = 2\log(1+e^{-2c})$. This is positive for finite c and tends to zero as $c \to \infty$. For every $c > 0$ both labels are correct. Thus the infimum is zero, with no finite minimizing slope along this perfectly separating dataset.

Marking: use the displayed subpart allocations; award credit for the logical or numerical intermediate steps.

Source: PDF pp.31-37; supplementary Hessian and nonattainment proof.

Section C. Proofs and Formula Derivations

28. Derive softmax normalization and the coupled learning rule [10 marks]

(a) [3 marks] Exponentiating yields $P_i = e^{a_i} P_K$. Summing probabilities gives $1 = P_K[1 + \sum_{i<K} e^{a_i}]$. Therefore $P_K = [1 + \sum_{i<K} e^{a_i}]^{-1}$ and $P_i = e^{a_i}/[1 + \sum_{j<K} e^{a_j}]$. This is softmax with reference score zero. Adding a common score to every class yields the symmetric form $y_i = e^{a_i}/\sum_j e^{a_j}$.

(b) [4 marks] Let $Z = \sum_k e^{a_k}$. The quotient rule gives $\partial y_i/\partial a_j = (\delta_{ij} e^{a_i} Z - e^{a_i} e^{a_j})/Z^2 = y_i(\delta_{ij} - y_j)$. Then

$$\frac{\partial \ell}{\partial a_j} = -\sum_i \frac{r_i}{y_i} y_i(\delta_{ij} - y_j) = -r_j + y_j \sum_i r_i = y_j - r_j.$$

This uses $\sum_i r_i = 1$. Thus $\partial \ell/\partial w_{j0} = y_j - r_j$ and $\nabla_{w_j} \ell = (y_j - r_j)x$, with sums over samples for the full loss.

(c) [3 marks] Adding $c(x)$ multiplies every numerator and the denominator by $e^{c(x)}$, which cancels. For K classes with d features and intercepts, fixing one class score to zero leaves $(K-1)(d+1)$ free coefficients instead of the redundant $K(d+1)$. A nontrue class has $r_j = 0$ but generally $y_j > 0$, so its error $y_j - r_j$ remains nonzero through the common probability normalization.

Marking: use the displayed subpart allocations; award credit for the logical or numerical intermediate steps.

Source: PDF pp.38-44; supplementary parameter-identifiability proof.

Section D. Applications

29. Audit one binary batch update and the resulting decision [12 marks]

(a) [4 marks] All probabilities are $1/2$. The intercept gradient is $1/2 - 1/2 - 1/2 = -1/2$. The slope gradient is $(1/2)(-2) + (-1/2)(1) + (-1/2)(3) = -3$. Hence $b_{\text{new}} = 0.05$ and $w_{\text{new}} = 0.3$.

(b) [4 marks] New logits are $(-0.55, 0.35, 0.95)$, with probabilities approximately $(0.3659, 0.5866, 0.7211)$. The loss is

$$\log(1 + e^{-0.55}) + \log(1 + e^{-0.35}) + \log(1 + e^{-0.95}) \approx 1.3158,$$

below the initial $3 \log 2 \approx 2.0794$. The first term uses label zero and a negative score; the other two use positive labels.

(c) [2 marks] The first probability is below $1/2$, and the other two are above it, so all three labels are correct. Positive correct-side logits can still increase in magnitude, raising correct-class probabilities and reducing cross-entropy; zero label error is not zero cross-entropy.

(d) [2 marks] Compare $4(1-y)$ with y: choose C_1 only if $y > 4/5$. None of the three fitted probabilities exceeds 0.8, so all are assigned to C_2. This changes the decision rule while retaining the same fitted probabilities.

Marking: use the displayed subpart allocations; award credit for the logical or numerical intermediate steps.

Source: PDF pp.31-37; new numerical application and cost-sensitive extension.

Section D. Applications

30. Train a three-class model and check the normalization [12 marks]

(a) [3 marks] The target is $r = (0, 0, 1)^\mathsf{T}$, all logits are zero, probabilities are $(1/3, 1/3, 1/3)$, and cross-entropy is $\log 3$.

(b) [5 marks] Each row is $(y_i - r_i)\widetilde{x}^\mathsf{T}$, so

$$\nabla_B \ell = \begin{bmatrix} 1/3 & 2/3 & -1/3 \\ 1/3 & 2/3 & -1/3 \\ -2/3 & -4/3 & 2/3 \end{bmatrix}, \qquad B_{\text{new}} = \begin{bmatrix} -1/6 & -1/3 & 1/6 \\ -1/6 & -1/3 & 1/6 \\ 1/3 & 2/3 & -1/3 \end{bmatrix}.$$

Both are 3×3. The signs are from subtracting one half of the gradient.

(c) [2 marks] Multiplying each row by $\widetilde{x}$ gives logits $(-1, -1, 2)$. Thus $y_3 = e^3/(2 + e^3) \approx 0.9094$, each other probability is $1/(2 + e^3)$, and loss is $\log(1 + 2e^{-3}) \approx 0.0949$. Class C_3 wins.

(d) [2 marks] The shared row adds ten to every logit, so all probabilities remain identical after cancellation. For stable evaluation, subtract the largest original logit, two, obtaining $(-3, -3, 0)$, and normalize the exponentials $(e^{-3}, e^{-3}, 1)$.

Marking: use the displayed subpart allocations; award credit for the logical or numerical intermediate steps.

Source: PDF pp.41-44; new numerical application.

Lecture Alignment and Coverage

Primary source: `Linear_Discrimination.pdf` in `E:\CS182\2026fall\lecture`. All PDF pages count the cover as page 1. Additional arguments are marked as supplementary derivations when appropriate. These are original practice questions, not copied homework questions.

T/F

1 PDF p.18.

2 PDF pp.11-15; supplementary degenerate case.

3 PDF pp.11,26,36; supplementary scale argument.

4 PDF pp.35,44.

5 PDF pp.26,36-37.

6 PDF p.43.

7 PDF pp.17-18,38-41; supplementary comparison.

8 PDF pp.40-41; supplementary identifiability argument.

9 PDF pp.4-6,22-31.

10 PDF pp.19-20.

11 PDF pp.29-31,47.

12 PDF pp.32,36-37; supplementary existence analysis.

Multiple response

13 PDF pp.32-35; supplementary sum/mean comparison.

14 PDF pp.17-18.

15 PDF pp.24,30; supplementary application of lecture 05 loss rule.

16 PDF p.37; supplementary connection to regularization.

17 PDF pp.26,33; supplementary maximum calculation.

18 PDF pp.31-34.

19 PDF pp.22-30,48.

20 PDF p.43.

21 PDF pp.8-9,48; supplementary example.

22 PDF pp.13-15; supplementary numerical example.

23 PDF pp.38-41.

24 PDF pp.38-44; supplementary matrix and identifiability notation.

Proofs and applications

25 PDF pp.11-15; supplementary projection and scale proof.

26 PDF pp.22-30.

27 PDF pp.31-37; supplementary Hessian and nonattainment proof.

28 PDF pp.38-44; supplementary parameter-identifiability proof.

29 PDF pp.31-37; new numerical application and cost-sensitive extension.

30 PDF pp.41-44; new numerical application.